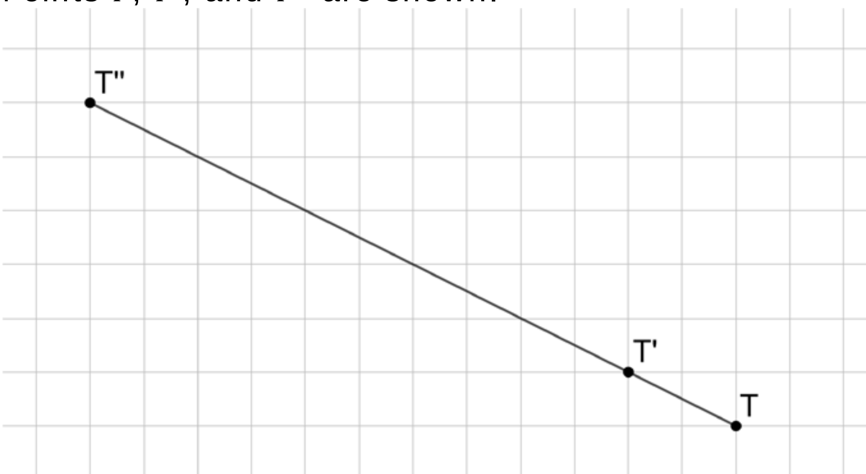
Geometry
Unit 1
Lesson 4 Practice

Name Answer Key________
Date___________________
Period__________________

Points $T$, $T'$, and $T''$ are shown.




1. Determine the number of copies of $\overline{TT'}$ that will fit on $\overline{TT''}$ . 6

2. Use the definition of dilation to explain why $\overline{TT''}$ is a dilation of $\overline{TT'}$.
$\overline{TT''}$ is a proportional increase of $\overline{TT'}$. It is 6 times as long as $\overline{TT'}$.

3. Complete the statement.

Point
- o $T$
- ✓ $T'$
- o $T''$

partitions
- o $\overline{TT'}$
- o $\overline{T'T''}$
- ✓ $\overline{TT''}$

so that the ratio of
- o $\frac{TT'}{T'T''}$
- ✓ $\frac{TT'}{TT''}$

is
- o $\frac{1}{5}$.
- ✓ $\frac{1}{6}$.

Use the Pythagorean theorem to complete the table.

| | Segment | Length |
|---|---|---|
| 4. | $\overline{TT'}$ | $2^2 + 1^2 = c^2$<br>$5 = c^2$<br>$\sqrt{5} = c$ |
| 5. | $\overline{T'T''}$ | $5^2 + 10^2 = c^2$<br>$125 = c^2$<br>$5\sqrt{5} = c$ |
| 6. | $\overline{TT''}$ | $6^2 + 12^2 = c^2$<br>$180 = c^2$<br>$6\sqrt{5} = c$ |

7. Complete the statements.

Point $T'$ is

- o $\frac{1}{5}$
- ☑ $\frac{1}{6}$

of the distance from $T$ to $T''$. The scale factor that can be used to multiply

- o $\overline{TT'}$
- o $\overline{T'T''}$
- ☑ $\overline{TT''}$

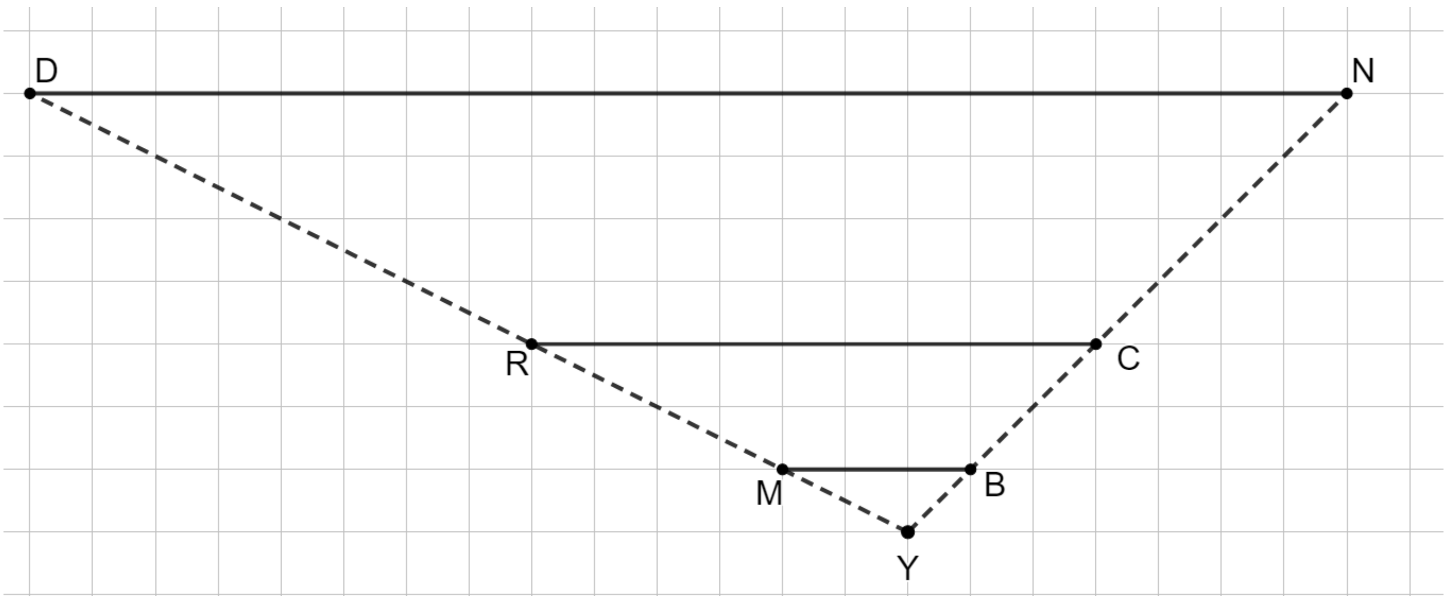
by to produce

- ☑ $\overline{TT'}$
- o $\overline{T'T''}$
- o $\overline{TT''}$

is

- o $\frac{1}{5}$.
- ☑ $\frac{1}{6}$.

Line segments $MB$, $RC$, and $DN$ are shown where $\overline{MB}$ is dilated using point $Y$ to create $\overline{RC}$ and $\overline{DN}$.




Determine each of the ratios.

| | Ratio | Length |
|---|---|---|
| 8. | $YM:YR$ | $\sqrt{5}:3\sqrt{5}$<br>$1:3$ |
| 9. | $MB:RC$ | $3:9$<br>$1:3$ |
| 10. | $MB:DN$ | $3:21$<br>$1:7$ |